

brainimg: A Reproducible Systems Study of Generative-Recall Image Compression

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Abstract

Classical image formats store pixels—either directly, as transform coefficients, or as latent codes—and are therefore resolution-bound and tied to the exact appearance of the original. This paper presents **brainimg**, a small prototype image format that stores the **meaning** of an image (a text caption) plus a tiny **structural blueprint** (128×128 depth, Canny-edge, and ADE20K segmentation maps) and a seed, and regenerates a visually faithful image on decode using one of fourteen pluggable diffusion decoder backends: Stable Diffusion 1.5 (default) or SDXL with two-to-three ControlNets, either of those plus ByteDance’s Hyper-SD 8-step distilled LoRA (turbo), Z-Image-Turbo with a single Union ControlNet, Qwen-Image (Apache 2.0) with InstantX’s Union ControlNet, HunyuanDiT v1.2 with separate depth + canny ControlNets, NVIDIA SANA 600M with an HED ControlNet fed the canny map, FLUX.2-klein-4B as an img2img pseudo-ControlNet fed the depth map, or FLUX.1-Depth-dev / FLUX.1-Canny-dev with channel-concat conditioning, optionally with Hyper-SD’s 8-step FLUX LoRA. We frame brainimg as a working, reproducible instantiation of the “Semantic-Relational Field / generative-recall” paradigm: rather than compressing appearance, it stores the scene’s semantics and geometry and lets a diffusion model repaint it. We describe the format schema, the four-stage encoder (VLM captioning, depth estimation, edge extraction, semantic segmentation), the fourteen decoder backends, the per-device memory/precision strategies, and the brightness/saturation post-processing with a gamma fallback for extreme targets. Using measurements reproducible from the committed repository on an AMD x86_64 CPU-only target with 188 GB RAM, we report blueprint sizes of 2.7–8.4 KB (compression ratios of 2.2×–124.8×), deterministic reconstruction given a fixed seed, and a per-backend fidelity and speed comparison on four classic USC-SIPI test images (mandril, peppers, cameraman, airplane) in which Hyper-SD 8-step distilled schedules **beat** their 30-step counterparts on SD 1.5 (+0.54 dB), and a ControlNet scale sweep finds that lower depth scales (0.8 vs 1.5) improve fidelity with the Depth-Anything-V2-Base stack. We are explicit that brainimg is lossy-by-design, decoder-dependent, and unsuited to forensic/medical use; we position it as a systems study of a novel paradigm rather than a replacement for JPEG.

1. Introduction

Every widely used image format answers a variant of the same question: *how do we store these pixels efficiently?* PNG stores exact colors and predicts the next pixel; JPEG/WebP discard transform coefficients the human visual system barely notices; AVIF generalizes this to AV1 intra-frames. All three paradigms

share two properties: the file is bound to a native resolution, and “loss” manifests as visible artefacts—blockiness, ringing, colour bleeding.

Neuroscience offers a different model. The retina compresses roughly 130 million photoreceptor signals into about one million optic-nerve fibres (a 130:1 ratio) by transmitting *edges and contrast* rather than absolute brightness. Deeper in the visual cortex, representation becomes increasingly semantic—neurons respond to faces, objects, materials—until recall itself becomes a generative act: to imagine an apple is not to “open a file” but to drive top-down signals from concept areas back into the visual cortex and *hallucinate* the sensory experience back into existence. The brain stores meaning and the rules to rebuild, not a pixel grid.

This paper is a reproducible systems study of a compression paradigm that mimics that idea. We ask:

Can a modern diffusion model serve as the decoder for a kilobyte-scale image format, conditioned on a text caption and a handful of low-resolution structural maps?

We do **not** claim a new state of the art, nor that the result replaces JPEG. We claim that the paradigm is implementable today, on commodity hardware (including an 8 GB Apple Silicon laptop), that it is deterministic given a seed, and that it exhibits qualitatively different—and in some ways useful—failure modes from transform codecs.

Contributions

1. **Paradigm framing.** We articulate the “Semantic-Relational Field / generative-recall” image-compression paradigm and map it onto a concrete encoder–format–decoder architecture.

1. **A versioned, forward-compatible file format.** `brainimg v0.1` is a small JSON document carrying a caption, three optional conditioning maps, a seed, and colour statistics. Optional fields allow older files to decode unchanged as new conditioners are added.

1. **A four-stage encoder** combining a vision-language model (Qwen2-VL on MLX, Qwen2.5-VL-7B on CPU/CUDA), Depth-Anything-V2-Base, OpenCV Canny, and OneFormer ADE20K semantic segmentation, with explicit memory release between stages so the pipeline fits in 8 GB.

1. **Fourteen pluggable decoder backends** spanning seven model families (SD 1.5, SDXL, Z-Image, Qwen-Image, HunyuanDiT, SANA, FLUX), each with an optional Hyper-SD 8-step distilled turbo variant for SD 1.5, SDXL, and FLUX. All backends consume the same blueprint; new backends require no schema change.

1. **Empirical findings on distilled schedules and scale tuning.** On the SIPI mandril test image, Hyper-SD’s 8-step distilled LoRA *beats* the 30-step non-turbo path on SD 1.5 (+0.54 dB) while running 3.1× faster on CPU. A grid sweep of ControlNet conditioning scales finds that Depth-Anything-V2-Base’s sharper depth map over-constrains at the

historical default of 1.5; lowering depth to 0.8 and raising seg to parity (1.0) yields a measurable lift on the turbo path.

1. **A gamma fallback for the brightness-clamp edge case.** When the uniform

gain clamp [0.5, 2.0] cannot reach an extreme target, a per-channel gamma curve (same exponent, clamped to [0.3, 3.0]) closes the residual gap without the clipping artefacts a $>2\times$ gain would cause.

The remainder of the paper is organised as follows. §2 surveys related work in classical and learned compression, text-to-image generation, and ControlNet conditioning. §3 describes the method (format, encoder, decoder, device strategy). §4 reports measurements reproducible from the committed repository. §5 discusses the paradigm’s properties—resolution independence, semantic loss, decoder dependency—and honest limitations. §6 concludes and points to planned work.

2. Related Work

2.1 Classical image codecs

The three dominant compression paradigms are *spatial/predictive* (PNG, run-length + DEFLATE over filtered scanlines), *spectral/transform* (JPEG’s discrete cosine transform, WebP’s VP8 intra, AVIF’s AV1 intra), and *fractal/wavelet* (JPEG 2000). All are pixel-bound: the file encodes a fixed grid at a fixed resolution, and “loss” is a transform artefact. brainimg departs from all of these by storing *no pixel data at all*.

2.2 Learned / neural compression

Neural compression typically maps pixels to a lower-dimensional latent space via an autoencoder and entropy-codes the latents. The decoder is a learned up-sampler. brainimg is adjacent but distinct: its “latent” is not a continuous vector but a *human-readable caption plus a few low-resolution maps and a seed*, and the decoder is a general-purpose text-to-image diffusion model conditioned by ControlNets rather than a codec-specific autoencoder. The decoder is therefore large (4 GB for SD 1.5) and shared across all files, while each file is tiny and self-describing.

2.3 Text-to-image diffusion and ControlNet

Stable Diffusion 1.5 [Rombach et al., 2022] performs latent diffusion conditioned on a CLIP text embedding. ControlNet [Zhang et al., 2023] adds a trainable copy of the UNet encoder blocks that forces the diffusion process to follow an external spatial conditioner (depth, edges, segmentation, pose). brainimg uses two ControlNets unconditionally (depth + Canny) and a third (ADE20K segmentation) when the blueprint carries a seg map. The combination of CLIP caption conditioning with multi-map ControlNet conditioning and a stored seed is, to our knowledge, novel as a *file-format* design, though the individual components are off-the-shelf.

2.4 Vision-language captioning

Accurate captions are the semantic backbone of the format. We use the Qwen2-VL-2B-Instruct 4-bit model via MLX on Apple Silicon for fast, low-memory captioning, and fall back to the Qwen2.5-VL-7B-Instruct model via

transformers elsewhere. A known failure mode (§4.6) is that the captioner can misidentify scene elements; because the conditioning maps drive structural fidelity, a wrong caption biases mood more than geometry.

2.5 The biological inspiration

The retina's 130:1 compression via lateral inhibition, sparse coding in V1, and top-down generative recall are well documented in the neuroscience literature. We use these only as *motivation*; brainimg is an engineering artefact, not a brain model.

3. Method

brainimg has three parts: an *encoder* that extracts a blueprint from a source image, a *file* that stores it, and a *decoder* that regenerates an image from the blueprint. Encoder and decoder run as **separate processes** so their heavy models are never resident simultaneously—an important constraint on the 8 GB Apple Silicon target.

```
Source image --> ENCODER --> .brainimg (JSON, ~3-10 KB) --> DECODER --> image
(jpg/png)      |                                     (SD+ControlNet) (any
size)

+- 1. Caption (Qwen-VLM)          --> prompt
+- 2. Depth   (Depth-Anything-V2) --> 128^2 JPEG
+- 3. Canny   (OpenCV)            --> 128^2 PNG
+- 4. Seg     (OneFormer ADE20K)  --> 128^2 PNG (optional)
```

3.1 The .brainimg file format

A .brainimg file is a small UTF-8 JSON document, typically 3–10 KB regardless of source resolution. The schema is versioned (`format_version` is currently "0.1"). Required fields are `format_version`, `original_width`, `original_height`, `prompt`, `depth_map_b64`, `canny_map_b64`, and `seed`. Optional fields, added after the initial release, are stored so that older files still decode: `segmentation_map_b64` enables the seg ControlNet only when present and non-empty. The extra dictionary absorbs unknown fields for forward compatibility. A representative file:

```
{
  "format_version": "0.1",
  "original_width": 1024, "original_height": 768,
  "prompt": "a red apple on a wooden table next to a window",
  "negative_prompt": "blurry, low quality, deformed, watermark, jpeg artifacts",
  "depth_map_b64": "<base64 128x128 JPEG>",
  "canny_map_b64": "<base64 128x128 PNG>",
  "segmentation_map_b64": "<base64 128x128 PNG, optional>",
  "seed": 42, "steps": 30,
  "target_brightness": 142.5,
  "target_saturation": 88.3,
  "extra": { "color_style": "vivid, saturated colors, natural lighting, red dominant tones" }
}
```

The format module (`brainimg/format.py`) is deliberately free of any ML import so the schema and round-trip tests run without downloading models. `MAP_SIZE` (currently 128) controls conditioning-map resolution; raising it

improves structural fidelity at the cost of a slightly larger file and requires re-encoding existing samples.

3.2 Encoder

The encoder (`brainimg/extract.py`) runs four extractors in sequence, releasing memory between heavy stages via `free_torch()` / `free_mlx()` so the pipeline fits in 8 GB.

Stage 1 — Captioning. A Qwen2-VL vision-language model is prompted with the instruction “*In one short sentence, describe this image for reconstruction: main objects, layout, lighting, and colors.*” On Apple Silicon the MLX 4-bit `mlx-community/Qwen2-VL-2B-Instruct-4bit` model is used (fast, low memory); on any other platform the transformers Qwen/Qwen2.5-VL-7B-Instruct model is used. Both produce an equivalent caption; the 7B is noticeably more detailed. MLX memory is released before any PyTorch stage runs.

Stage 2 — Depth map. Depth-Anything-V2-Base (`depth-anything/Depth-Anything-V2-Base-hf`) via a HuggingFace depth-estimation pipeline produces a grayscale depth map (near = bright, far = dark), downsampled to `MAP_SIZE × MAP_SIZE` with Lanczos resampling and JPEG-encoded at quality 70.

Stage 3 — Canny edges. OpenCV’s Canny detector (`low=50, high=150`) is run on the grayscale image downsampled with `INTER_AREA`, then PNG-encoded to keep edges crisp.

Stage 4 — Segmentation (optional). OneFormer ADE20K (`shi-labs/oneformer_ade20k_swin_tiny`) produces a semantic map that is colourised with the exact 150-class ADE20K palette the SD 1.5 seg ControlNet was trained on, then nearest-neighbour downsampled to `MAP_SIZE × MAP_SIZE` and PNG-encoded. This stage was added after the initial v0.1 release; older files omit it and decode unchanged.

Colour-style and statistics. Alongside the four conditioners the encoder computes two scalar statistics of the original image, stored for decoder-side post-processing. *Brightness* is the mean Rec.609 luminance on a 0–255 scale:

$$B = \text{mean}(0.299 R + 0.587 G + 0.114 B).$$

Saturation is the mean normalised channel spread, scaled to 0–255 (0 = grey, 255 = fully saturated):

$$S = 255 \cdot \text{mean}\left(\frac{\max_c - \min_c}{\max(\max_c, 1)}\right).$$

A short *colour-style prefix* is also derived (e.g. “*dark, low-key lighting, red dominant tones*”) and stored in `extra`; the decoder prepends it to the caption only when the combined length fits the CLIP 77-token limit, so the caption itself is never truncated.

3.3 Decoder

The decoder (`brainimg/generate.py`) regenerates an image from the blueprint using one of fourteen pluggable backends, all sharing the same blueprint schema and the same brightness/saturation post-processing:

--model	Base	Conditioning	Steps (default)	License
sd15 (default)	stable-diffusion-v1-5	depth + canny (+ seg) ControlNets	30	CreativeML Open RAIL-M
sd15-turbo	SD 1.5 + Hyper-SD 8-step LoRA	depth + canny (+ seg) ControlNets	8	CreativeML Open RAIL-M
sdxl	stable-diffusion-xl-base-1.0	depth + canny (+ seg) ControlNets	30	CreativeML Open RAIL-M
sdxl-turbo	SDXL + Hyper-SD 8-step LoRA	depth + canny (+ seg) ControlNets	8	CreativeML Open RAIL-M
zimage	Tongyi-MAI/Z-Image-Turbo	single Union ControlNet (depth)	9 (8-step Turbo)	Tongyi-MAI non-commercial
qwen-image	Qwen/Qwen-Image	single Union ControlNet (depth)	50	Apache 2.0
hunyuan	Tencent-Hunyuan/HunyuanDiT-v1.2-Diffusers-Distilled	depth + canny ControlNets	25	tencent-hunyuan-community
sana	Efficient-Large-Model/Sana_600M_1024px	single HED ControlNet (canny map)	20	MIT
flux2-klein	black-forest-labs/FLUX.2-klein-4B	img2img (depth map as starting image)	4	Apache 2.0
flux-depth	black-forest-labs/FLUX.1-Depth-dev	channel-concat depth_map_b64	30	FLUX.1-dev non-commercial
flux-canny	black-forest-labs/FLUX.1-Canny-dev	channel-concat canny_map_b64	30	FLUX.1-dev non-commercial
flux-depth-turbo	FLUX.1-Depth-dev + Hyper-SD 8-step FLUX LoRA	channel-concat depth_map_b64	8	FLUX.1-dev non-commercial
flux-union	black-forest-labs/FLUX.1-dev + Shakker-Labs Union ControlNet	Union ControlNet (depth + canny)	24	FLUX.1-dev non-commercial
sd35	stabilityai/stable-diffusion-3.5-large	depth + canny ControlNets	50	stabilityai-ai-community

All fourteen backends consume the **same blueprint**; the schema is unchanged when a new decoder is added. Two additional CLI variants of the counted

backends are supported by `decoder.py` but not enumerated as separate paper backends: `--model hunyuan-full` (the non-distilled 50-step HunyuanDiT v1.2, used in §4.6 to isolate the distillation artefact) and `--model \flux-canny-turbo` (Hyper-SD 8-step LoRA on FLUX.1-Canny-dev, the canny-channel counterpart of `flux-depth-turbo`). We describe the SD 1.5 / SDXL path in detail (the historical default), then summarise the turbo, Z-Image, Qwen-Image, HunyuanDiT, SANA, FLUX.2-klein, and FLUX paths that diverge structurally.

3.3.1 SD 1.5 / SDXL (ControlNet stack)

The default backend. SD 1.5 ships at 512², SDXL at 1024².

Conditioning. The depth and Canny maps are decoded from base64 and upsampled to the target size (Lanczos for depth, nearest-neighbour for Canny and seg to keep edges/palette crisp). When a segmentation map is present it is appended as a third conditioner. Default conditioning scales for SD 1.5 are depth 0.8, Canny 1.0, seg 1.0 — tuned via a grid sweep on the SIPI samples (§4.8). The historical defaults (depth 1.5, canny 1.2, seg 0.9) were set for the older Depth-Anything-Small + no-seg pipeline; Depth-Anything-V2-Base’s sharper depth map over-constrains at 1.5, and the ADE20K seg ControlNet adds material cues that warrant parity with Canny.

VAE substitution. The stock SD 1.5 VAE is replaced with the fine-tuned `stabilityai/sd-vae-ft-mse` for cleaner decode, better skin tones and colours, and fewer washed-out highlights, at negligible runtime cost.

Scheduler and steps. The pipeline uses a `UniPCMultistepScheduler` with a default of 30 inference steps (raised from 20). ControlNet scales and classifier-free guidance (default 7.5) are CLI-tunable.

Determinism. A `torch.Generator` seeded with `data.seed` makes generation deterministic: re-decoding with the same seed reproduces the same image exactly (verified to produce 0 pixel difference between runs).

Colour post-processing. SD 1.5 systematically over-brightens and over-saturates. The decoder corrects this against the stored targets in two ordered steps, chosen so each step does not undo the other:

1. *Saturation first*, by scaling the HSV-S channel by

$\text{target_saturation} / \text{current_saturation}$, iterated 1–2× to converge (HSV-S is not linear in the metric above). Ratios are clamped to $[0.5, 2.0]$ to avoid clipping artefacts. This perturbs brightness, which the next step corrects.

1. *Brightness last*, by a uniform RGB gain $\text{target_brightness} / \text{current_brightness}$. A uniform gain preserves colour balance and leaves

the saturation metric (a ratio of channels) invariant, so correcting brightness last does not undo the saturation work. The same $[0.5, 2.0]$ clamp applies.

The step is a no-op for older files whose targets are 0.0, or when the generation’s stats are already within 2 % of the targets. When the clamped gain cannot reach an extreme target (e.g. darkening 210→80 needs ratio 0.38, clamped to 0.5 → 105), a **gamma fallback** applies a per-channel gamma curve

arr^γ (same exponent on every channel, clamped to $[0.3, 3.0]$) to close the residual gap. Gamma preserves colour balance approximately (not exactly like a uniform gain, but far better than clipping) and reaches extreme targets without the posterisation a $>2\times$ gain would cause. The gamma exponent is $\gamma = \log(t/255) / \log(c/255)$ where c and t are the current and target brightnesses normalised to $[0, 1]$.

Style-prefix gating. The stored colour-style prefix is prepended to the caption only when the combined length fits within the CLIP 77-token limit, biasing mood without ever truncating the caption. Older files with no stored style use the caption verbatim. (For Z-Image-Turbo and FLUX, which use a Qwen / T5 text encoder with a 512-token limit, the style prefix is prepended unconditionally.)

3.3.2 Z-Image-Turbo (Union ControlNet)

--model zimage swaps the SD stack for **Tongyi-MAI/Z-Image-Turbo** (a 6 B-parameter single-stream DiT distilled for 8 steps) plus the alibaba-pai/Z-Image-Turbo-Fun-Controlnet-Union-2.1 *Union* ControlNet (the full 2.1-8steps variant, 6.4 GB). The Union net is a single network that takes one conditioning image per call and bakes the conditioning *type* (depth/canny/pose/mlsd/hed) into its training; we feed the blueprint's depth_map_b64 because depth carries the most structural information. The blueprint's canny_map_b64 and segmentation_map_b64 are **silently ignored** on this path—no schema change. Z-Image runs in **bf16 throughout**, which sidesteps the MPS fp16 NaN bug entirely (a different dtype, no int8 quantization is used). Defaults: guidance_scale = 0.0 (Turbo is distilled for zero CFG), 9 inference steps (8 DiT forward passes on the Turbo schedule), 1024 max side.

3.3.3 FLUX.1-Depth-dev / FLUX.1-Canny-dev (channel-concat)

--model flux-depth and --model flux-canny load Black Forest Labs' **FLUX.1** guidance-distilled Control variants. FLUX.1 is a 12 B-parameter MMDiT transformer with a T5-XXL text encoder (CLIP-L is the small auxiliary encoder). The FLUX.1-*-dev Control checkpoints bake the conditioning into the transformer via **channel-wise concatenation** of the conditioning image—not a separate ControlNet model. Diffusers' FluxControlPipeline takes a single control_image per call; we feed depth_map_b64 for flux-depth and canny_map_b64 for flux-canny. The other map and any segmentation_map_b64 are silently ignored. bf16 throughout (FLUX's native dtype, sidesteps the MPS fp16 NaN bug). Defaults: cfg 10.0 (depth) or 30.0 (canny), 30 inference steps, 1024 max side, max_sequence_length = 512 (T5-XXL). No per-call controlnet_conditioning_scale—the conditioning strength is fixed by the trained checkpoint, so the --depth-scale / --canny-scale CLI flags are accepted for argument parity but silently ignored on FLUX paths. **License:** FLUX.1-Depth-dev / -Canny-dev carry the FLUX.1-dev non-commercial license.

3.3.4 Hyper-SD turbo distillation (SD 1.5 / SDXL / FLUX)

ByteDance's **Hyper-SD** [Ren et al., 2024] applies trajectory-segmented consistency distillation to produce small (70–150 MB) LoRAs that fold the

SD 1.5, SDXL, and FLUX.1-dev base models down to **8 inference steps** while preserving the stock ControlNets / channel-concat conditioning. The LoRA is loaded via `pipe.load_lora_weights` and fused with `pipe.fuse_lora(0.125)` (per the model card) after device placement, so the UNet / transformer weights are permanently adjusted — no per-call LoRA overhead. For SD 1.5 and SDXL, the scheduler is swapped to `DDIMScheduler(timestep_spacing="trailing")`; for FLUX, no scheduler swap is needed (`FlowMatchEulerDiscreteScheduler` works natively with fewer steps). The turbo paths ignore the file's stored step count (tuned for 20–30 step SD schedules) and use 8 steps unless `--steps` is passed.

FLUX turbo compatibility fix. The Hyper-SD FLUX LoRA was trained on the base FLUX.1-dev, not the Control variants. The Control transformer's `x_embedder` (patch embedding) has extra input channels for the control image (128 vs 64) and a `context_embedder` that base dev doesn't have, so those LoRA deltas are shape-incompatible. The decoder strips the `x_embedder` and `context_embedder` keys (and the `transformer.prefix`, which diffusers adds internally) before loading. The attention/FFN LoRA deltas — the bulk of the distillation — load cleanly and are what matters for the 8-step schedule. Guidance drops to 3.5 (the FLUX dev default, not the 10.0 / 30.0 the non-turbo control variants use).

3.3.5 Qwen-Image (Union ControlNet, Apache 2.0)

`--model qwen-image` loads Alibaba's **Qwen-Image** [Wu et al., 2025] (arXiv 2508.02324, Apache 2.0) — a DiT with a Qwen2.5-VL text encoder (512-token limit) — plus InstantX's Qwen-Image-ControlNet-Union, a single Union ControlNet supporting canny + depth + pose + soft-edge. Same pattern as Z-Image: depth-only conditioning, blueprint's canny/seg ignored, bf16 throughout. Defaults: 50 steps, `true_cfg_scale = 4.0` (Qwen-Image's CFG parameter, distinct from `guidance_scale`), `controlnet_conditioning_scale = 0.9`, 1024 max side. The Qwen-Image backend is the only fully-open (Apache 2.0) high-quality option in the stack; FLUX is non-commercial and Z-Image is non-commercial.

3.3.6 HunyuanDiT v1.2 (depth + canny ControlNets)

`--model hunyuan` loads Tencent's **Hunyuan-DiT** [Li et al., 2024] (arXiv 2405.08748) v1.2 Distilled — a bilingual (Chinese + English) DiT with **two separate ControlNets** (depth + canny), the same two-conditioner pattern as SD 1.5 / SDXL rather than a Union net. The blueprint's seg map is **silently ignored** (no seg ControlNet exists for HunyuanDiT). HunyuanDiT only supports fixed resolutions (1024, 1280, ...); on a 512-side blueprint it auto-upscals to 1024. Text encoding uses BERT + T5 (the CLIP 77-token limit does not apply; the colour-style prefix is prepended unconditionally). bf16 throughout (HunyuanDiT's native dtype, sidesteps the MPS fp16 NaN bug). Defaults: depth scale 0.8, canny scale 0.8, `guidance_scale = 3.0`, 25 steps (distilled), 1024 max side. `--model hunyuan-full` swaps the Distilled base for the non-distilled HunyuanDiT-v1.2-Diffusers (50 steps, otherwise identical) so we can isolate whether distillation is the source of the visual artefacts (§4.6). **License:** tencent-hunyuan-community.

3.3.7 FLUX.1-dev Union ControlNet (depth + canny)

--model flux-union loads Black Forest Labs' **FLUX.1-dev** base plus the **Shakker-Labs/FLUX.1-dev-ControlNet-Union-Pro** checkpoint [Shakker Labs & InstantX, 2024]. The Union ControlNet bundles seven conditioning modes (canny, tile, depth, blur, pose, gray, low-quality) into a single network; we feed **depth (mode 2) and canny (mode 0) simultaneously** via FluxMultiControlNetModel and control_mode=[2, 0]. The blueprint's seg map is **silently ignored** (no seg mode). guidance_scale = 3.5, 24 steps, controlnet_conditioning_scale = 0.4 per map, 1024 max side, max_sequence_length = 512. --quantize FP8-quantizes the transformer and T5-XXL (weights only), dropping resident CPU RAM from 24 GB to 12 GB. **License:** FLUX.1-dev non-commercial (gated).

3.3.8 Stable Diffusion 3.5 Large (depth + canny ControlNets)

--model sd35 loads Stability AI's **Stable Diffusion 3.5 Large** [Esser et al., 2024] (8B MMDiT) with the official **depth** and **canny** ControlNets. Both ControlNets are 8B checkpoints; we load them separately and wrap them in SD3MultiControlNetModel so depth and canny are fed together. Text encoding uses CLIP-L, CLIP-G, and T5-XXL (the SD3 text encoder stack). bf16 throughout. Defaults: depth scale 0.7, canny scale 0.7, guidance_scale = 4.5, 50 steps, 1024 max side, max_sequence_length = 256. The seg map is **silently ignored** (no SD3.5 seg ControlNet). **License:** stabilityai-ai-community (gated, free for non-commercial and for commercial use up to \$1M annual revenue).

3.3.9 SANA 600M (HED ControlNet, MIT)

--model sana loads NVIDIA's **SANA** [Xie et al., 2024] (arXiv 2410.10629, MIT license) — a 600 M-parameter linear DiT with a DC-AE VAE (32× spatial compression) and a T5 text encoder. diffusers ships SanaControlNetPipeline + SanaControlNetModel since 0.38. **Only an HED (soft-edge) ControlNet exists for SANA** — no depth or canny ControlNet has been trained. We feed the blueprint's **canny** map to the HED ControlNet as the closest available conditioning type. This is a type mismatch: HED produces soft probability edges while Canny produces hard binary edges, creating a PSNR-vs-colour trade-off (§4.6, §4.7). The blueprint's depth and seg maps are **silently ignored**. The diffusers-format ControlNet is a community conversion (ishan24/Sana_600M_1024px_ControlNet_diffusers) of the official NVlabs checkpoint; the base model is the diffusers port from Efficient-Large-Model. bf16 throughout. Defaults: HED scale 0.4 (tuned via sweep on the SIPI samples at 1024² — §4.8), guidance_scale = 4.5, 20 steps, 1024 max side, max_sequence_length = 300. SANA is the fastest 1024-native backend (54 s at 1024², 5 GB RAM) but the lowest-PSNR backend due to the HED/canny mismatch.

3.3.10 FLUX.2-klein-4B (img2img pseudo-ControlNet, Apache 2.0)

--model flux2-klein loads Black Forest Labs' **FLUX.2-klein-4B** (Apache 2.0, ungated, 4 B-parameter rectified-flow transformer, 4-step distilled) as an **image-to-image** model. FLUX.2-klein has **no ControlNet**; we use it as a pseudo-ControlNet by feeding the blueprint's **depth map** as the image parameter (the starting point for img2img). The

model encodes the depth map into latents and denoises from there with the caption as the text guide, “editing” the depth map into a photorealistic image rather than being structurally constrained by a ControlNet. The blueprint’s canny and seg maps are **silently ignored** (only one image input). bf16 throughout. Defaults: `guidance_scale = 1.0`, 4 steps, 1024 max side, `max_sequence_length = 512`. The only real FLUX.2 ControlNet (alibaba-pai Union, depth + canny) requires the VideoX-Fun library + the 32 B gated FLUX.2-dev, which is impractical on CPU; the `img2img` path is the practical fallback.

3.4 Device and precision strategy

The current dev target is an **AMD x86_64 CPU-only box with 188 GB RAM** (no CUDA/MPS), so the CPU fp32 path is the primary measurement platform. The decoder also supports Apple Silicon (MPS int8) and NVIDIA CUDA (fp16), but those are secondary. On MPS, SD 1.5 fp16 matmuls produce NaNs (a black output frame); the decoder therefore int8-quantizes the UNet and all ControlNets (weights *and* activations) via optimum-quanto to avoid fp16 matmuls entirely, while the VAE runs in fp32 for a clean final decode. On CPU, full fp32 is supported (best fidelity, slow) and an optional int8 weights mode fits in 5 GB at a small quality cost. On CUDA, fp16 works correctly and is fast. Z-Image, Qwen-Image, HunyuanDiT, SANA, FLUX.2-klein, FLUX, and SD3.5 are bf16-native and sidestep the MPS fp16 NaN bug entirely; on a CPU-only box they are kept resident in host RAM (diffusers’ `enable_model_cpu_offload` raises `RuntimeError` without an accelerator to offload to).

--device / --model	Precision	RAM	Speed	Fidelity
cpu (sd15, default target)	fp32 (no quant)	10 GB	156 s @ 512 ²	best (sd15)
cpu --model sd15-turbo	fp32 + Hyper-SD 8-step LoRA	10 GB	51 s @ 512 ²	good (+0.54 dB vs 30-step)
cpu --model sdxl	fp32 (no quant)	17 GB	989 s @ 512 ²	best (sdxl)
cpu --model sdxl-turbo	fp32 + Hyper-SD 8-step LoRA	17 GB	76 s @ 512 ²	good (−2.58 dB vs 30-step)
cpu --model zimage	bf16 resident	18 GB	308 s @ 512 ²	good (depth-only)
cpu --model qwen-image	bf16 resident	20 GB	1006 s @ 512 ²	good (depth-only)
cpu --model hunyuan	bf16 resident	12 GB	912 s @ 1024 ²	good PSNR, poor visual (§4.6)
cpu --model sana	bf16 resident	5 GB	54 s @ 1024 ²	lowest PSNR (HED/canny mismatch)
cpu --model flux2-klein	bf16 resident	13 GB	42 s @ 512 ²	good PSNR, poor colour
cpu --model flux-depth --quantize	bf16 + FP8 (host RAM)	12 GB	510 s @ 512 ²	best (FLUX)

--device / --model	Precision	RAM	Speed	Fidelity
cpu --model flux-depth-turbo --quantize	bf16 + FP8 + Hyper-SD 8-step	12 GB	475 s @ 512²	within noise of 30- step
cpu --model flux-union -- quantize	bf16 + FP8 (host RAM)	12 GB	860 s @ 512 ²	mid PSNR
cpu --model sd35	bf16 resident	16-20 GB	3100 s @ 1024 ² →512 ²	mid PSNR, can zoom at 512 ²
mps (sd15, Apple Silicon)	int8 weights + activations	5 GB	medium (8 GB Mac)	fair
cuda (sd15)	fp16	5 GB	fast	good

The encoder/decoder device selection is centralised in `brainimg/device.py`, which also provides `free_torch()` / `free_mlx()` memory-release helpers.

4. Experiments

All measurements below are reproducible from the committed repository; we use only data already present in the repo and do not run the heavy ML models here. Where a number is documented in the project (`README.md`, `TODO.md`) we cite it; where it is an on-disk file size we measured it directly with `stat`. We do not report PSNR/SSIM/LPIPS/FID because those runs are out of scope for this writing session; §5 flags the planned evaluation.

4.1 Setup

- **Hardware:** AMD x86_64 CPU-only, 188 GB RAM (the current dev target),

32 cores, 16 torch threads. All measurements in §4 are from this machine.

The M1/8 GB Apple Silicon results in `README.md` (59 s @ 256²) are historical and noted where relevant.

- **Software:** Python 3.12, uv-managed environment, `requirements.txt`.

`diffusers` 0.38, `peft` 0.19 (for LoRA loading), `optimum-quanto` (FP8).

- **Models:** SD 1.5 + `sd-vae-ft-mse` + 3 ControlNets (default), SDXL,

Hyper-SD 8-step LoRAs (SD 1.5, SDXL, FLUX), Z-Image-Turbo, Qwen-Image, HunyuanDiT v1.2 Distilled + full, SANA 600M, FLUX.2-klein-4B, FLUX.1-Depth-dev / -Canny-dev. Captioner is transformers Qwen2.5-VL-7B (CPU fallback; MLX is Apple-Silicon-only).

- **Samples:** `samples/real.jpg` (256×256 puppy JPEG, 13,430 B),

`samples/mandril_color.tif` (512×512 SIPI mandril, 787,420 B),

`samples/peppers_color.tif` (512×512 SIPI peppers, 786,572 B),

`samples/cameraman.tif` (512×512 SIPI cameraman, grayscale, 262,750 B),

`samples/airplane.tif` (512×512 SIPI F-16 fighter jet, 786,572 B),

`samples/test512.jpg` (512×512, 49,690 B). Blueprints: `real.brainimg`,

`mandril.brainimg`, `peppers.brainimg`, `cameraman.brainimg`,

`airplane.brainimg`, `test512.brainimg`. Mandril decodes use seed 200,

peppers seed 300, cameraman seed 100, airplane seed 400, all at 512×512

output unless noted. The four SIPI images are public-domain standard test images from the USC-SIPI database, used here in place of the historically

common Lenna image (retired from many venues for reasons documented at <https://en.wikipedia.org/wiki/Lenna>).

4.2 Compression

Table 2 reports blueprint sizes and compression ratios against the source files. The “5.0×” figure for the puppy in `README.md` corresponds to an earlier pre-segmentation v0.1 file of 2.7 KB; the committed `real.brainimg` includes the seg map and colour statistics and is 6,120 B (2.2×), reflecting the richer current schema.

Source file	Source size	.brainimg size	Ratio
<code>samples/real.jpg</code> (puppy, 256 ²)	13,430 B	6,120 B (<code>real.brainimg</code>)	2.2×
<code>samples/test512.jpg</code> (512 ²)	49,690 B	5,777 B (<code>test512.brainimg</code>)	8.6×
<code>samples/cameraman.tif</code> (512 ² , grayscale)	262,750 B	7,693 B (<code>cameraman.brainimg</code>)	34.2×
<code>samples/mandril_color.tif</code> (512 ²)	787,420 B	8,386 B (<code>mandril.brainimg</code>)	93.9×
<code>samples/peppers_color.tif</code> (512 ²)	786,572 B	7,652 B (<code>peppers.brainimg</code>)	102.8×
<code>samples/airplane.tif</code> (512 ²)	786,572 B	6,305 B (<code>airplane.brainimg</code>)	124.8×
(documented) <code>samples/real.jpg</code>	13.4 KB	2.7 KB (pre-seg v0.1)	5.0×

Two properties are worth noting. First, the blueprint size is **roughly constant** (a few KB) regardless of source resolution or colour depth: a 256² puppy, a 512² grayscale cameraman, and a 512² colour mandril all produce 6–8 KB files, because the stored maps are fixed at 128². Second, the compression ratio therefore *grows with source size*: the large uncompressed TIFFs compress 94–125× while the already-compressed puppy JPEG compresses 2×. This is the opposite of transform codecs, whose ratios are largely independent of whether the source is raw or pre-compressed, and is a direct consequence of storing meaning rather than pixels.

4.3 Reconstruction quality (qualitative)

Reconstruction is semantically faithful (same scene, layout, and lighting) but not pixel-identical—by design. The committed repository ships one side-by-side figure:

- `comparison.jpg` — original vs. brainimg reconstruction of `samples/real.jpg`.

Per `README.md`, the captioner correctly described the scene (“a black puppy sitting on a wooden surface”) and the decoder produced a visually faithful reconstruction at 256×256 in 59 s on the M1/8 GB machine.

Across the SIPI subjects the per-backend character is best read from the fidelity numbers in §4.7: SDXL at its native 1024² reproduces the mandril’s broad palette most faithfully; SD 1.5 turbo gives the best speed/fidelity trade-off on the same image; FLUX.1-Depth-dev at 512² stays close to the source palette where SDXL@512 drifts (§4.6); HunyuanDiT and FLUX.2-klein score in the top half by PSNR yet are visually weak due to palette collapse/shift (§4.6, §4.7). Cross-subject, the grayscale cameraman is the easiest to match (FLUX depth turbo reaches 15.80 dB) and the F-16’s clean lines suit the canny ControlNet (SD 1.5 turbo reaches 15.05 dB on `airplane.brainimg`, the highest PSNR of any SIPI subject at 512²).

Side-by-side grids of all backends on each SIPI subject can be regenerated with `scripts/make_backend_grid.py <subject>` (mandril, peppers, cameraman, airplane); the script writes `<subject>_grid.jpg` next to the source. The per-backend reconstruction PNGs and `<subject>_<backend>\_comparison.jpg` side-by-sides are similarly produced by `decoder.py` and `scripts/make_comparison.py`. These artifacts are deliberately gitignored (see `.gitignore`: they are large decoder outputs, not source) and regenerated on demand.

4.4 Determinism

`README.md` reports that re-running the decoder with the same seed reproduces the same image exactly, verified as 0 pixel difference between runs. The SDXL mandril run was additionally verified to produce an identical md5 across runs. Determinism is a property of the format: the seed is stored in the file, so a `.brainimg` file plus a fixed decoder yields a bit-identical image.

4.5 Device and precision ablation

Table 1 (§3.4) summarises the supported operating modes. The key engineering finding is that on Apple Silicon the SD 1.5 naive fp16 path is unusable (NaNs), forcing int8 weights + activations; this halves memory but degrades structural fidelity relative to CPU fp32. CPU fp32 is the recommended SD 1.5 mode on a high-RAM machine for best quality; CUDA fp16 is the recommended mode for speed. Z-Image-Turbo, Qwen-Image, HunyuanDiT, SANA, FLUX.2-klein, and FLUX are bf16 throughout (their native dtype), which sidesteps the MPS fp16 NaN bug entirely; FLUX additionally supports FP8 quantization of the transformer + T5 via `optimum.quanto` to halve resident memory at a small quality cost.

4.6 Known failure modes

Five failure modes remain documented in `TODO.md`:

- **Captioner misidentification.** The 7B captioner can misidentify scene

elements. On the SIPI mandril it correctly identified the colourful face but on some dark subjects has read hair as a hat or accessories. Because the conditioning maps (depth/Canny/seg) capture the true structure regardless, a wrong caption biases mood more than geometry—but it does bias generation. Mitigations noted for future work: a larger VLM or caption ensembling.

- **SDXL hue-distribution drift at small sizes.** When SDXL is decoded at

512×512 on a source whose palette spans multiple hue bands (e.g. the mandril’s blue face stripes + orange/red nose + green/yellow fur), the output can collapse the blue band: the mandril source is 30.7% blue/purple pixels, SDXL@512 drops it to 7.6%, while SDXL-turbo@512 overshoots to 49.7%. This is a content/palette drift, not a stat drift: a single global HSV-H rotation aligns means but cannot reshape distributions, and a rotation large enough to chase a different distribution recolours neutrals and skin in a way that reads worse than the original drift.

Workaround: prefer SDXL at 1024×1024, where the drift is much smaller.

FLUX.1-Depth-dev at 512×512 produces a less-drifted palette than SDXL@512 on the same blueprint (§4.7).

- **HunyuanDiT pixel-metric-vs-perceptual disconnect.** HunyuanDiT scores

10.68 dB PSNR (mid-pack, §4.7) and is visually the worst backend by a wide margin — visible artefacts and a blue/purple band collapse (8.8% vs the mandril source’s 30.7%) that MSE/PSNR do not capture. The good MSE relative to SD 1.5 likely comes from getting overall brightness/layout right at 1024² while producing texture/feature artefacts. The likely cause is a language mismatch: HunyuanDiT is bilingual with a BERT tokenizer trained primarily on Chinese data; the English caption may produce inferior results regardless of tuning. Not recommended for visual use; kept for the systems-study comparison.

- **SANA HED/canny mismatch.** SANA’s only available ControlNet is an HED

(soft-edge) net; we feed the blueprint’s canny (hard binary edge) map to it. This type mismatch creates a PSNR-vs-colour trade-off: on the mandril at 1024² the default scale 0.4 gives 7.69 dB — the lowest PSNR of any backend — while preserving the blue band at 28.5% (close to the source’s 30.7%). Raising the scale would trade colour fidelity for PSNR. SANA is the fastest 1024-native backend (54 s at 1024², 5 GB RAM) but the lowest-PSNR backend due to the mismatch.

- **FLUX.2-klein img2img palette shift.** The pseudo-ControlNet img2img

path (§3.3.10) reaches #2 PSNR overall (11.01 dB at 512²) but shifts the colour palette: 41.9% blue/purple vs the mandril source’s 30.7%, and on the peppers (12% blue source) the FLUX depth turbo path collapses to 0.3% blue — the model converts the depth map’s grayscale into warm tones regardless of the caption. As with HunyuanDiT, this is a case where pixel-level MSE rewards overall brightness/layout accuracy while missing a colour-distribution failure that the human eye sees immediately.

The brightness-clamp edge case that was previously listed here has been **fixed** with a gamma fallback (§3.3.1): when the uniform gain clamp [0.5, 2.0] cannot reach an extreme target, a per-channel gamma curve closes the residual gap. Three new tests in `tests/test_color.py` cover the gamma darkening, brightening, and approximate ratio preservation paths.

4.7 Per-backend fidelity and speed comparison (mandril, 512×512)

We report three pixel-level metrics (MSE, PSNR, MAE) and wall time for each decoder backend on the SIPI mandril blueprint (`samples/mandril_color.tif`, 512×512, seed 200), all decoded at 512×512 on the AMD CPU target (188 GB RAM) unless noted. Metrics are computed by `scripts/compare_backends.py` against the source resized to 512×512. Two cross-subject sanity rows (peppers, cameraman) at the best-performing backend confirm the codec works across palettes and grayscale.

Backend	Steps	Time (s)	MSE ↓	PSNR (dB) ↑	MAE ↓
SD 1.5 (tuned scales)	30	156	8696	8.74	75.6
SD 1.5 turbo (Hyper-SD, tuned)	8	51	7683	9.28	70.3

Backend	Steps	Time (s)	MSE ↓	PSNR (dB) ↑	MAE ↓
SDXL	30	989	3253	13.01	46.1
SDXL turbo (Hyper-SD)	8	76	5891	10.43	60.4
Z-Image (depth-only)	8	308	7023	9.67	67.6
Qwen-Image (depth-only)	50	1006	5944	10.39	61.5
SANA (HED/canny, scale 0.4)	20	54 @ 1024 ²	11067	7.69	88.1
HunyuanDiT (depth+canny, 1024 ²)	25	912	5559	10.68	59.8
FLUX.2-klein (img2img, 512 ²)	4	42	5159	11.01	57.2
FLUX.1-Depth-dev (FP8)	30	510	6620	9.92	64.6
FLUX.1-Depth-dev turbo (FP8)	8	475	6648	9.90	64.5
FLUX Union (depth+canny, FP8)	24	860	7817	9.20	72.2
SD 3.5 (depth+canny, 1024 ² →512 ²)	50	3100	8048	9.07	72.8

| **Cross-subject sanity (FLUX depth turbo, FP8)** | | | | |

| peppers_color.tif (512²) | 8 | 187 | 4142 | 11.96 | 51.5 |

| cameraman.tif (512² grayscale) | 8 | 207 | 1712 | **15.80** | 28.4 |

Seven observations:

1. Hyper-SD 8-step distilled schedules beat their 30-step counterparts on

SD 1.5. The turbo path scores 9.28 dB vs 8.74 dB for the 30-step path

(+0.54 dB) — at 3.1× less wall time. On FLUX, however, the turbo path (9.90 dB) does *not* beat the 30-step path (9.92 dB) on this image — within noise. The distilled-schedule-wins finding is SD 1.5-specific on this subject; it held strongly on the retired Lenna sample for both SD and FLUX (see §A) but does not generalise to the mandril’s broad palette.

1. SDXL is the best result across all backends (13.01 dB, 989 s) — the

30-step SDXL at its native 1024² resolution beats every other backend on the mandril. FLUX.2-klein img2img (11.01 dB, 42 s) follows at #2 by PSNR, and Qwen-Image (10.39 dB) at #3 — both ahead of the FLUX depth path. The ranking differs from the retired Lenna sample (§A), where FLUX depth turbo was #1 — a reminder that single-image rankings do not generalise.

1. ControlNet scale tuning on SD 1.5.

The tuned defaults (depth 0.8, canny 1.0, seg 1.0) reflect the §4.8 scale sweep, which found depth 0.6 marginally better than 0.8 on the mandril (9.37 vs 9.28 dB) — the 0.8/1.0/1.0 compromise is retained because it is robust across both broad-palette (mandril) and narrow-palette subjects.

1. Qwen-Image (Apache 2.0) is competitive with SDXL turbo (10.39 dB

vs 10.43 dB) despite being depth-only (no canny/seg), but slow at 50 steps (1006 s). It beats Z-Image (9.67 dB) on the same depth-only pattern. The Apache 2.0 license is a practical advantage over FLUX’s non-commercial license for distribution.

1. Z-Image is the weakest depth-only backend (9.67 dB, 308 s) —

slightly below SDXL turbo (10.43 dB, 76 s) and Qwen-Image (10.39 dB). SANA (7.69 dB, 54 s @ 1024²) is the lowest-PSNR backend on this subject due to the HED/canny mismatch (§4.6), though it is the fastest 1024-native backend.

1. **Pixel-level PSNR does not track perceptual quality for two backends.**

HunyuanDiT (10.68 dB, mid-pack) is visually the worst backend by a wide margin — visible artefacts and a blue/purple band collapse (8.8% vs source 30.7%) — and FLUX.2-klein (#2 by PSNR, 11.01 dB) shifts the palette (41.9% blue vs source 30.7%). MSE rewards getting overall brightness/layout right but does not penalise texture/feature or colour-distribution failures; this is the concrete pixel-metric-vs-perceptual disconnect flagged in §5.3 as motivation for adding LPIPS / CLIP-score / FID in a real evaluation.

1. **The two newest ControlNet backends sit in the mid-PSNR range.**

FLUX Union (9.20 dB, 860 s) and SD 3.5 (9.07 dB, 3100 s) both underperform the headline FLUX depth turbo and SDXL results on the mandril. SD 3.5 additionally requires native 1024² generation: when forced to 512² it produces a zoomed/cropped composition, so the decoder generates at 1024² and downscales. Both are included as additional ControlNet options rather than top-tier fidelity choices.

These numbers do not generalise beyond the SIPI samples (four sources with varied palettes); §5.3 notes that a real evaluation requires more subjects and perceptual metrics (LPIPS, CLIP-score, FID). Side-by-side grids of all backends on each SIPI subject can be regenerated with `scripts/make_backend_grid.py` (mandril, peppers, cameraman, airplane).

4.8 ControlNet scale sweep

The historical SD 1.5 conditioning defaults (depth 1.5, canny 1.2, seg 0.9, cfg 7.5) were set for the older Depth-Anything-Small + no-seg pipeline. With Depth-Anything-V2-Base (sharper depth) and the ADE20K seg ControlNet now in the stack, we ran a grid sweep on `samples/mandril_color.tif` at 512×512 with `sd15-turbo` (8 steps, seed from the blueprint), using `scripts/sweep_scales.py` — 10 configurations, loading the pipeline once and varying only the scale/cfg pair. (The original sweep also covered `samples/test512.jpg`; see §A for those archived numbers.)

Findings:

- **Lower depth helps.** Depth 0.6 (9.37 dB) beats 0.8 (9.28 dB) beats 1.0 (9.05 dB) on the mandril. The V2 depth map is sharper than the Small model's, so high scales over-constrain and fight the caption.
- **Seg at parity (1.0) is the compromise.** The ADE20K seg ControlNet adds material cues that were missing in the old no-seg pipeline; parity with canny (1.0) is robust.
- **Canny 1.0 beats 1.2.** A small but consistent improvement.

New SD 1.5 defaults: depth 0.8, canny 1.0, seg 1.0, cfg 7.5 (was 1.5/1.2/0.9/7.5). The 0.8 compromise is retained: 0.6 wins on the mandril by 0.09 dB but the older sweep found 0.6 hurt the narrower-palette test512

sample, and 0.8 is robust across both broad and narrow palettes. SDXL defaults (1.0/0.8/0.6) were left unchanged — they were already in the good region.

4.9 MAP_SIZE regression

A TODO item proposed raising MAP_SIZE from 128 to 256 for sharper conditioning maps. We tested this on a 512×512 sample and it **regressed on every backend**: SD 1.5 30-step −0.65 dB, SD 1.5 turbo −0.85 dB, SDXL turbo −0.57 dB (archived numbers, see §A). File size also grew 2.5× (8 KB → 20 KB). The ControlNets appear over-constrained by the sharper maps at 512×512 output — 128 maps upscaled 4× to 512 give the right amount of structural grip, while 256 maps upscaled 2× over-specify edges/depth. MAP_SIZE stays at 128; the finding is documented in TODO.md and format.py. Re-evaluation is warranted if the default output size moves to 1024.

5. Discussion

5.1 Properties of the paradigm

Resolution independence. A .brainimg file has no native resolution: it is a set of instructions for a generative model, so the same few-KB file can be decoded at 256, 512, 1024, or higher, the decoder simply generating more pixels. This is qualitatively different from transform codecs, where up-sampling a small file produces a blurry large image.

Semantic, not pixel, loss. In JPEG, loss is block artefacts and colour bleeding. In brainimg, “loss” means the model may render a slightly different wood grain or fabric pattern than the original, while the fabric still looks photorealistic. The lost data is micro-detail, replaced by synthesised micro-detail. This is desirable for human-visual consumption and unacceptable for forensic/medical/legal use where exact pixels matter.

Editability. Because the file is a self-describing JSON blueprint, a user can edit the caption, mood prefix, or conditioning scales and re-decode to relight or re-pose the scene—an early hint of the “edit the recipe, not the pixels” property of the Semantic-Relational Field paradigm.

Determinism. Stored seed + fixed decoder ⇒ bit-identical re-decode. This addresses the usual “hallucination is non-deterministic” objection to generative codecs: within a fixed decoder version, a file *is* a stable image.

5.2 Limitations

- **Decoder dependency.** Unlike JPEG’s kilobyte decoder, brainimg’s decoder is a multi-gigabyte neural network (anywhere from 4 GB for SD 1.5 to 22 GB for FLUX.1 bf16, reducible to 12 GB with FP8 quantization). A device without the standard decoder cannot view the image—the codec is more like a video codec in this respect. On the other hand, the decoder is *shared* across all files, so the per-file cost remains a few KB.
- **Compute cost.** Decompression runs a diffusion model: 51 s for SD 1.5

turbo at 512² on the AMD CPU target, 475 s for FLUX turbo (FP8), 17 min for Qwen-Image at 50 steps. The turbo paths make interactive use plausible on CPU; the 30-step paths remain in the minutes-per-image range. This still precludes gallery scrolling or video use cases.

- **Quality depends on device.** CPU fp32 gives the best SD 1.5

reconstruction; on 8 GB Apple Silicon, int8 quantization (required by the MPS fp16 NaN bug) degrades structural fidelity. Z-Image-Turbo, Qwen-Image, HunyuanDiT, SANA, FLUX.2-klein, FLUX, and SD3.5 are bf16-native and are not supported on 8 GB Apple Silicon; FLUX CPU requires 22 GB host RAM (or 12 GB with --quantize), while SD3.5 requires 16-20 GB. SANA is the lightest of the bf16-native backends (5 GB RAM).

- **Lossy by design.** Reconstruction is semantically faithful, not

pixel-identical. The format is explicitly unsuited to medical, legal, or forensic images.

- **Captioner accuracy.** Wrong captions bias generation (§4.6). Structure

is the primary fidelity driver, which limits but does not eliminate the impact of caption errors.

- **Per-backend palette fidelity.** SDXL and Z-Image, run at sizes smaller

than their native training resolution (1024), can drift into a different hue *distribution* than the source (§4.6). HunyuanDiT and FLUX.2-klein exhibit a more severe form of this — a palette *collapse/shift* (HunyuanDiT 8.8% blue vs mandril source 30.7%; FLUX.2-klein 41.9% blue) that even MSE does not catch (§4.6, §4.7). On the peppers (12% blue source), the FLUX depth turbo path collapses to 0.3% blue. This is a content/palette drift the existing brightness/saturation post-processor cannot correct. Prefer each backend's native resolution (512 for SD 1.5, 1024 for SDXL / Z-Image / HunyuanDiT / SANA / FLUX.2-klein / FLUX) when colour fidelity matters.

- **License footprint.** The default backend (SD 1.5 + ControlNets) is

CreativeML Open RAIL-M. SDXL is the same. Z-Image-Turbo is non-commercial. HunyuanDiT is tencent-hunyuan-community (a bespoke community license, not OSI-approved). FLUX.1-Depth-dev / -Canny-dev are FLUX.1-dev non-commercial; SD3.5 is stabilityai-ai-community (gated, free up to \$1M annual revenue). Qwen-Image, SANA, and FLUX.2-klein-4B are **Apache 2.0** — the three fully-open options. If distribution matters, the SD 1.5/SDXL, Qwen-Image, SANA, and FLUX.2-klein paths are the permissively licensed choices.

5.3 Planned evaluation (not run here)

A complete evaluation would include: PSNR/SSIM/LPIPS against originals for the sample images across all backends and device modes; CLIP-score as a semantic-fidelity proxy; FID against a small natural-image set; a bitrate-matched comparison to JPEG/WebP/AVIF at similar file sizes; and an ablation removing each ControlNet in turn to quantify each conditioner's contribution. The current measurements are on four SIPI test images (mandril, peppers, cameraman, airplane) with varied palettes; a real

evaluation requires more subjects. The repository is structured to support these via `encoder.py`/`decoder.py` and the `samples/` directory.

6. Conclusion

brainimg is a small, reproducible prototype of a different way to compress images: store the *meaning and structure* of a scene and let a diffusion model repaint it. We have described the format schema, the four-stage encoder, the fourteen decoder backends (SD 1.5, SDXL, their Hyper-SD turbo variants, Z-Image-Turbo, Qwen-Image, HunyuanDiT v1.2, SANA, FLUX.2-klein-4B img2img, FLUX.1-Depth-dev / -Canny-dev with Hyper-SD turbo on the depth variant, FLUX.1-dev + Shakker-Labs Union ControlNet, and Stable Diffusion 3.5 Large) with their quality post-processing and gamma brightness fallback, and the device/precision tradeoffs across CPU fp32, MPS int8, and CUDA fp16. Using measurements reproducible from the committed repository on an AMD CPU target with 188 GB RAM, we observe few-kilobyte blueprints ($2.2\times$ – $124.8\times$ compression), deterministic reconstruction given a seed, and two empirical findings: (1) Hyper-SD’s 8-step distilled schedules *beat* their 30-step counterparts on SD 1.5 (+0.54 dB) while running $3.1\times$ faster on CPU, and (2) lower ControlNet depth scales (0.8 vs 1.5) improve fidelity with the Depth-Anything-V2-Base stack. SDXL (13.01 dB PSNR, 989 s) is the best result across all backends on the mandril. We also report a concrete pixel-metric-vs-perceptual disconnect: HunyuanDiT and FLUX.2-klein score in the top half by PSNR yet are visually weak backends due to palette collapse/shift that MSE does not capture. We frame this as a systems study of a paradigm, not a JPEG replacement, and we are explicit about its limitations—decoder dependency, compute cost, per-backend palette fidelity, and lossy-by-design semantics.

Planned work tracked in `TODD.md` includes per-region hue transfer (a fix for the SDXL hue-distribution drift documented in §4.6) and a captioner upgrade (§4.6). The `MAP_SIZE` 128→256 bump, ControlNet scale tuning, brightness-clamp gamma fix, and all turbo distillation backends are done.

7. Reproducibility

The project is self-contained and engineered to run on commodity hardware.

Setup (Python 3.12 via uv):

```
uv venv -p 3.12
source .venv/bin/activate
uv pip install -r requirements.txt
# On non-Apple platforms, uninstall the non-functional MLX stub wheels:
pip uninstall -y mlx mlx-vlm mlx-lm
```

Encode and decode:

```
# photo -> tiny .brainimg blueprint
python encoder.py samples/real.jpg -o out.brainimg --seed 42

# blueprint -> regenerated image (CPU fp32, best fidelity)
python decoder.py out.brainimg -o recon.png --device cpu
```

```
# CUDA fp16 (fast), or CPU int8 (low RAM)
python decoder.py out.brainimg -o recon.png --device cuda
python decoder.py out.brainimg -o recon.png --device cpu --quantize
```

The format and colour post-processing modules are deliberately ML-free and can be tested without downloading models:

```
pytest # tests/test_format.py + tests/test_color.py + tests/
test_flux_config.py, runs in seconds
```

Known gotchas (from AGENTS.md): the MPS fp16 NaN bug is *not* a bug to “fix” by switching to fp16—int8 quantization is the intended workaround (Apple Silicon only; not relevant on the AMD CPU target). The Hyper-SD FLUX LoRA was trained on base FLUX.1-dev, not the Control variants; the decoder strips the shape-incompatible `x_embedder` / `context_embedder` deltas before loading (§3.3.4). Re-decoding with the same seed reproduces the image exactly.

Appendix A. Archived measurements (retired Lenna sample)

The measurements in §4.7–§4.9 were originally conducted on the Lenna test image (`samples/lenna.tiff`, 512×512, seed 916570520). Lenna was removed from the sample set prior to publication per the well-known objection documented at <https://en.wikipedia.org/wiki/Lenna> the numbers below are retained for provenance and to document how the headline findings changed (or did not) when the sample set moved to the four USC-SIPI images.

A.1 Per-backend fidelity and speed (Lenna, 512×512)

Backend	Steps	Time (s)	MSE ↓	PSNR (dB) ↑	MAE ↓
SD 1.5 (old scales)	30	180	8763	8.70	77.5
SD 1.5 (tuned scales)	30	156	7560	9.35	70.8
SD 1.5 turbo (Hyper-SD, tuned)	8	50	7055	9.65	68.1
SDXL	30	220	5774	10.52	58.8
SDXL turbo (Hyper-SD)	8	69	6085	10.29	61.1
Z-Image (depth-only)	8	237	7651	9.29	70.3
Qwen-Image (depth-only)	50	1436	6810	9.80	68.4
SANA (HED/canny, scale 0.4)	20	52 @ 1024 ²	6633	9.91	64.2
HunyuanDiT (depth+canny, 1024 ²)	25	1004	2977	13.39	44.3
FLUX.2-klein (img2img, 512 ²)	4	240	2736	13.76	41.9
FLUX.1-Depth-dev (FP8)	30	654	3202	13.08	43.6
FLUX.1-Depth-dev turbo (FP8)	8	166	2314	14.49	37.1

On Lenna, FLUX.1-Depth-dev turbo was the best result (14.49 dB) and the distilled schedule beat the 30-step path on both SD 1.5 (+0.95 dB) and FLUX (+1.41 dB). On the mandril (§4.7), SDXL is best (13.01 dB) and the FLUX distilled schedule does *not* beat the 30-step path. The distilled-schedule-wins finding is therefore image-dependent, not a universal property of the distillation.

A.2 ControlNet scale sweep (Lenna + test512)

The original sweep covered `samples/lenna.tiff` and `samples/test512.jpg`.

Findings: depth 1.0 beats 1.5 on both; 0.8 beats 1.0; 0.6 beats 0.8 on

test512 but hurt Lenna. Seg 1.2 beat 0.9 on Lenna; 0.9 beat 1.2 on test512.

The 0.8/1.0/1.0 compromise was chosen as robust across both. The mandril sweep (§4.8) confirms 0.6 edges out 0.8 by 0.09 dB, but 0.8 is retained for cross-sample robustness.

A.3 MAP_SIZE regression (Lenna)

The 128→256 MAP_SIZE bump regressed on every backend on Lenna at 512×512:

SD 1.5 30-step -0.65 dB (8763 → 10185 MSE), SD 1.5 turbo -0.85 dB

(7934 → 9640), SDXL turbo -0.57 dB (6085 → 6928). File size grew 2.5×

(7.9 KB → 19.7 KB).

References

The references below are to canonical works and projects used by brainimg.

They are given in a lightweight author-year-title form suitable for a

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abovzv/sdxl_segmentation_controlnet_ade20k (SDXL stack).

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(the full 2.1-8steps variant; the *lite* 2601/2602 files are rejected by diffusers 0.38 due to a widened control_all_x_embedder input dim).

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Hugging Face; FLUX.1-dev non-commercial license. Underlies the FLUX Control variants and the flux-union backend.

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